📊 Data-Driven Problem Framing

Week 1 • Day 5 • Customer-Centric Foundations

Technical Product Manager Academy

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- Build an **evidence brief** that pairs every claim with its tier and source
- Use AI to pressure-test — not author — the problem statement
- Present a **10-minute integrated Week 1 artifact** to the full cohort
- Set up the bridge to Week 2 (metrics + strategic design)

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Review Week 1 artifact stack
09:15 – 10:30	Block 1 + Activity 1	Problem statement template
10:45 – 12:00	Block 2 + Activity 2	Evidence brief & AI critique
13:00 – 14:15	Block 3 + Activity 3	Rehearsal + peer dry-run
14:30 – 16:00	Block 4 + Readouts + Wrap	Full-cohort presentations

Block 1 — The Problem Statement

Scoped, verb-led, evidence-backed, falsifiable

📌 The Template

Who — [persona role + context]

Currently — [observed behavior, with frequency]

Which causes — [measurable consequence]

Because — [root cause, from 5 Whys]

We know this because — [evidence, tagged]

Success looks like — [the change we'd see if the problem is solved]

> The last line is the one most problem statements skip. Without it, there's no way to know when you're done.

📄 A FieldPulse Example

Who — Lead Dispatchers at 50–200 tech HVAC companies

Currently — Spend an average of 45 minutes post-shift reconciling paper tickets, truck stock, and timecards (observed, 6 of 8 ride-alongs)

Which causes — ~3.75 unpaid hours per dispatcher per week and a measured 22% turnover in the role (reported, HR data from 3 customers)

Because — Ticket submission requires connectivity techs don't have at the last job, so paper becomes the path of least resistance (5 Whys root)

We know this because — **Observed** 6 ride-alongs; **Reported** 4 interviews; **Inferred** turnover causality

Success looks like — Post-shift reconcile under 10 minutes for 80% of dispatchers within 6 months

🔴 Failure Modes of Problem Statements

Failure	Sounds like	Why it's weak
Solution in disguise	"Dispatchers need a mobile reconcile screen"	Pre-decides the how; not falsifiable
Consequence-free	"Dispatchers are frustrated with reconciliation"	No measurable cost; no way to prioritize
No root cause	"The process is inefficient"	Symptom, not mechanism
Ungrounded	"Customers tell us…"	No evidence tier; can't be audited
Success-agnostic	Missing the last line	Can't recognize solved

👥 Activity 1 — Draft Your Statement

Triads • 45 minutes

Using your Top 3 pain set and persona, draft a problem statement in the six-line template.

- Start with the strongest pain from Day 4
- Every line must survive the failure-mode table
- Mark which evidence tier each claim uses

⏱️ 20 min draft · 15 min failure-mode audit · 10 min peer-within-triad read

Block 2 — The Evidence Brief

What goes with the problem statement

📝 The Brief, One Page

> **The discipline:** Every claim in the problem statement appears in the evidence log. Nothing floats unsupported.

📝 The Brief, One Page

1. **The persona** — one paragraph from your Day 3 canvas

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1. **The persona** — one paragraph from your Day 3 canvas
2. **The pain map** — your Day 4 matrix, with Top 3 highlighted

> **The discipline:** Every claim in the problem statement appears in the evidence log. Nothing floats unsupported.

📝 The Brief, One Page

1. **The persona** — one paragraph from your Day 3 canvas
2. **The pain map** — your Day 4 matrix, with Top 3 highlighted
3. **The problem statement** — from Activity 1

> **The discipline:** Every claim in the problem statement appears in the evidence log. Nothing floats unsupported.

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1. **The persona** — one paragraph from your Day 3 canvas
2. **The pain map** — your Day 4 matrix, with Top 3 highlighted
3. **The problem statement** — from Activity 1
4. **The evidence log** — each claim: source, date, tier, who captured it

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1. **The persona** — one paragraph from your Day 3 canvas
2. **The pain map** — your Day 4 matrix, with Top 3 highlighted
3. **The problem statement** — from Activity 1
4. **The evidence log** — each claim: source, date, tier, who captured it
5. **The open questions** — what you don't yet know and plan to learn

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🧠 AI as Pressure-Tester

```
Role: Senior engineering lead reading a problem statement I intend to scope against.
Context: [paste problem statement + evidence log]
Constraints:
- Do not rewrite. Ask the questions you would ask me in a scoping meeting.
- Rank your questions by how much they'd change your estimate if unanswered.
- Flag any claim that is inferred or AI-generated that you'd want upgraded first.
Format: Numbered list of questions, ranked.
```

> **Rule:** AI can challenge the statement. AI does not get to write it. Ownership stays with the triad.

👥 Activity 2 — Pressure Test

Triads • 45 minutes

Run the pressure-test prompt. Then, as a triad:

- Answer every question you can from existing evidence
- Add unresolved questions to "Open questions" in the brief
- Upgrade or remove any claim that failed

⏱️ 10 min run prompt · 25 min respond · 10 min revise brief

Block 3 — Rehearsal

10 minutes, the right 10 minutes

🎤 The Readout Structure

> **Norm:** Don't defend in the Q&A. Listen and note. You'll act on the feedback in Week 2 pre-work.

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1. **0:00 – 1:30** Persona (who, observed behavior, 1 quote)

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1. **0:00 – 1:30** Persona (who, observed behavior, 1 quote)
2. **1:30 – 3:30** Top 3 pains (matrix, one slide)

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2. **1:30 – 3:30** Top 3 pains (matrix, one slide)
3. **3:30 – 5:30** Problem statement (read it, one line at a time)

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4. **5:30 – 7:30** Evidence sample + open questions

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4. **5:30 – 7:30** Evidence sample + open questions
5. **7:30 – 10:00** Q&A from room, questions only

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👥 Activity 3 — Paired Dry-Run

Paired triads • 45 minutes

Triads pair. Each does a **full timed 10-minute readout** to the other, with stopwatch.

- Partner writes down the **3 questions** they would ask in real Q&A
- Partner writes down **1 sentence** of what they heard the problem to be
- Compare that sentence to what the triad intended to convey

⏱️ 10 min each direction · 25 min debrief

Block 4 — Cohort Readouts

Week 1 integrated artifact

🏆 How We Score (Reminder)

Dimension	Weight
Customer authenticity	25%
Pain specificity	25%
Evidence quality	20%
Problem statement clarity	20%
AI-use transparency	10%

Each triad gets a score sheet from every peer triad + the instructor. Scores plus one strength and one suggestion.

👥 Week 1 Readouts

Full cohort • 10 min per triad + 3 min score-sheet writing

Order is random (draw triad names). Between triads, a 3-minute quiet block for written feedback.

- Presenting triad: follow the rehearsal structure
- Audience triads: questions only, no defending
- Everyone: written score + 1 strength + 1 suggestion

⏱️ Plan ~80–90 min for 6 triads

🎉 Week 1 Wrap

What we built (five artifacts)

1. Prompt Pattern Library (triad)
2. PR/FAQ
3. Validated persona canvas
4. Ranked pain-point matrix
5. Problem statement + evidence brief

What Week 2 does with them

- Problem statement → North Star metric
- Pain map → operational signals & KPIs
- Persona → customer journey map
- PR/FAQ → UX-principled re-framing

Weekend pre-read for Week 2: Re-read your problem statement. Bring 3 candidate metrics. Come curious about which one is the North Star.

💬 Closing Reflection

One sentence: what's the mindset you're taking into Week 2?